LEVEL UP ACADEMY

Class 10 | GSEB | Science

Pre-Board Mock 2 (GSEB pattern)

Chapter / Topic: Full syllabus

Name: _____ Roll No.: _____ Date: _____

Time: 3 hours **Max. Marks: 80**

General instructions:

1. Section A (Q1–24) is compulsory: 1 mark each.
2. Section B: answer any 9 of 12 questions (2 marks each). Section C: answer any 6 of 8 (3 marks each). Section D: answer any 5 of 7 (4 marks each).
3. Write balanced equations and draw neat, labelled diagrams wherever needed. Show formulae and units in numericals.
4. This mock follows the reported GSEB blueprint (A 24 · B 18 · C 18 · D 20). Check the latest official GSEB blueprint for changes.

SECTION A (Q1–8): CHOOSE THE CORRECT OPTION — 8 marks

Q1. An iron nail is kept in copper sulphate solution for some time. The colour of the solution becomes: **[1]**

(a) dark blue (b) colourless
(c) light green (d) red

Q2. Which of the following is an acidic oxide? **[1]**

(a) CaO (b) Na_2O
(c) MgO (d) SO_2

Q3. Vinegar is a dilute solution of: **[1]**

(a) ethanoic acid (b) methanoic acid
(c) citric acid (d) tartaric acid

Q4. Which of the following shows parasitic nutrition? **[1]**

(a) Amoeba (b) Cuscuta
(c) Mushroom (d) Grass

Q5. Which plant hormone inhibits growth? **[1]**

(a) Auxin (b) Gibberellin
(c) Cytokinin (d) Abscisic acid

Q6. A convex lens is also called a: **[1]**

(a) converging lens (b) diverging lens
(c) plane lens (d) cylindrical lens

Q7. Which instrument is used to measure electric current? **[1]**

(a) Voltmeter (b) Ammeter
(c) Galvanometer (d) Rheostat

Q8. Which of the following is biodegradable? **[1]**

(a) Plastic bag (b) Aluminium foil
(c) Vegetable peel (d) Glass bottle

SECTION A (Q9–14): FILL IN THE BLANKS — 6 marks

Q9. Coating iron with a thin layer of zinc is called _____. **[1]**

Q10. The chemical name of baking soda is _____. **[1]**

Q11. The SI unit of electric charge is the _____. **[1]**

Q12. The _____ is the main thinking part of the brain. **[1]**

Q13. The male reproductive part of a flower is the _____. **[1]**

Q14. The SI unit of resistance is the _____. **[1]**

SECTION A (Q15–19): STATE TRUE OR FALSE

5 marks

Q15. Ethanol turns blue litmus red. **[1]**

Q16. Arteries carry blood away from the heart. **[1]**

Q17. There is more variation in asexual reproduction than in sexual reproduction. **[1]**

Q18. Red light is scattered the most by particles in the atmosphere. **[1]**

Q19. The earth wire in domestic circuits has green insulation. **[1]**

SECTION A (Q20–24): ANSWER IN ONE WORD

5 marks

Q20. Name the acid present in the human stomach. **[1]**

Q21. Name the green pigment that absorbs light energy in photosynthesis. **[1]**

Q22. Which gas is produced when a reactive metal reacts with dilute hydrochloric acid? **[1]**

Q23. Which type of lens is used to correct myopia? **[1]**

Q24. What is the potential difference between the live and neutral wires in Indian homes? **[1]**

SECTION B

answer any 9 of 12 · 18 marks

Answer any 9 questions. 2 marks each.

Q25. Why is silver chloride stored in dark-coloured bottles? Write the equation. **[2]**

Q26. Give two uses of baking soda and two uses of washing soda. **[2]**

Q27. Arrange in decreasing order of reactivity: Cu, Zn, Fe, Mg, Ag. Which of these will not react with dilute HCl? **[2]**

Q28. Why is the conversion of ethanol to ethanoic acid called an oxidation reaction? **[2]**

Q29. What is the role of saliva in digestion? **[2]**

Q30. Why can amphibians and many reptiles manage with a three-chambered heart? **[2]**

Q31. How does chemical coordination take place in animals? **[2]**

Q32. What is binary fission? Give one example. **[2]**

Q33. Why does a child resemble its parents, but not exactly? **[2]**

Q34. Why is a concave mirror used in the headlights of vehicles? **[2]**

Q35. Resistors of 10 Ω and 15 Ω are connected in series to a 50 V supply. Find the current, and the potential difference across the 10 Ω resistor. **[2]**

Q36. What is a food web? How is it different from a food chain? **[2]**

SECTION C

answer any 6 of 8 · 18 marks

Answer any 6 questions. 3 marks each.

Q37. Balance the following equations and state the type of each reaction: **[3]**

(i) $H_2 + Cl_2 \rightarrow HCl$ **[1]**

(ii) $Na + H_2O \rightarrow NaOH + H_2$ **[1]**

(iii) $BaCl_2 + H_2SO_4 \rightarrow BaSO_4 + HCl$ **[1]**

Q38. Show the formation of sodium chloride and magnesium chloride by the transfer of electrons. (Na: 11, Mg: 12, Cl: 17) **[3]**

Q39. Write balanced equations for: (i) ethanoic acid + sodium hydroxide, (ii) burning of ethanol in air, (iii) methane + chlorine in sunlight. **[3]**

Q40. What are the different ways in which glucose is broken down to release energy in different organisms? **[3]**

Q41. How do the following plant hormones affect plants: (i) auxin, (ii) gibberellin, (iii) abscisic acid? **[3]**

Q42. Explain, with one example each: (i) fragmentation, (ii) budding, (iii) spore formation. **[3]**

Q43. For a convex lens, give the position, size and nature of the image when the object is (i) at infinity, (ii) at $2F_1$, (iii) between F_1 and the optical centre. **[3]**

Q44. An electric iron is rated 1500 W, 220 V. Find (i) the current it draws, (ii) its resistance, (iii) the heat it produces in 1 minute. **[3]**

SECTION D

answer any 5 of 7 · 20 marks

Answer any 5 questions. 4 marks each.

Q45. Why are metals high in the reactivity series (sodium, magnesium) extracted by electrolysis and not by reduction with carbon? Explain how sodium is obtained. **[4]**

Q46. (a) What does the pH of a solution tell us? [1]
(b) A milkman adds a very small amount of baking soda to fresh milk. (i) Why does he shift the pH of the milk from 6 to slightly alkaline? (ii) Why does this milk take a long time to set as curd? [3] **[4]**

Q47. Name the organs of the human excretory system and give the function of each. Name the functional unit of the kidney. **[4]**

Q48. (a) Name the main organs of the human female reproductive system and give one function of each. [3]
(b) What happens if the egg is not fertilised? [1] **[4]**

Q49. (a) What is dispersion of light? [1]
(b) Why does the sky look dark to astronauts in space? [1]
(c) Why does the Sun appear reddish at sunrise and sunset? [2] **[4]**

Q50. (a) State Joule's law of heating. [1]
(b) Give two practical applications of the heating effect of current. [1]
(c) Why is a fuse wire made of a material with a low melting point, and why is it connected in series with the live wire? [2] **[4]**

Q51. (a) What is biological magnification? [1]
(b) Why is its effect greatest in human beings? [1]
(c) Explain the 10 per cent law of energy flow with an example. [2] **[4]**

~ All the best for your boards! ~